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When do group incentives for retail store managers work?

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Abstract

Retail stores are geographically dispersed as a part of a multiunit organization. In such a setting, store managers play an important role in driving store performance. To motivate them to exert effort, retailers have provided group incentives for store managers. Using data from 75 stores of a U.S.-based retail chain that changed its incentive plan for store managers from being purely dependent on store performance to being dependent upon both store and corporate performance, we investigate the effect of this change on store performance. We also examine the moderating role of geographical proximity among stores and past performance of the focal store. To establish a causality, we identify a control group from a matched store of another retail chain in the same industry and perform a difference-in-differences analysis. We find that making managers' bonus dependent on corporate goals has resulted in overall worsening of store performance. However, such an effect is contingent upon both geographical proximity among stores and past performance of the focal store. The impact of the incentive change on store performance is *positively* moderated by the increase in stores within close geographical proximity, while it is *negatively* moderated by the past performance of the focal store. This paper broadens our understanding of how incentives work in practice, and more importantly, documents how firms can design more effective incentive schemes for managers by considering relevant conditions, such as stores' geographical networks and their past performance.

KEYWORDS
empirical, geographical proximity, incentive, retail operations

1 | INTRODUCTION

Retail organizations are complex multiunit enterprises made up of geographically dispersed stores. As such, store managers play a critical role in driving store performance—differences across store managers, for instance, explain about 27–35% of the variation in store-level performance (Siebert & Zubanov, 2010). Thus, it is vital to design appropriate incentives for store managers to motivate them to exert the effort required to improve retailers' financial performance (DeHoratius & Raman, 2007). In practice, we observe some retailers tie store managers' incentives to group performance in addition to their own store's performance and others to only their store's performance. For example, retailers such as Kroger[1] and Home Depot[2] tie store managers' compensation to both store and corporate performance while other retailers such as JCPenney and Tweeter Consumer Electronics tie store managers' compensation to only their store performance.

Such group incentives that tie an individual's bonus to higher level goals (e.g., corporate performance) are widely prevalent. Brown and Armstrong (1999), for instance, claim that more than 50% of major companies in the United States and Europe utilize some type of group-based compensation plan. Prior studies that examine the applicability of such group incentives typically investigate team members where collaborative opportunities among individuals exist due to geographical proximity. For example, Friebel et al. (2017) study sales associates collaborating in a retail store and Hamilton et al. (2003) examine plant workers collaborating in a garment factory. The efficacy of group incentives for retail store managers, however, is less clear because geographically dispersed stores may have limited opportunities to collaborate. So, we first examine the overall effect of the adoption of group incentives on store performance.

Geographical proximity among facilities (e.g., stores) has been an important consideration for many firms due to its impact on the firm performance. Prior literature has shown a positive outcome when entities are in geographical proximity

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because short distances facilitate frequent interactions, fostering knowledge transfer. For example, Bray et al. (2019) find much faster quality improvement for geographically gathered supply chains and Chu et al. (2019) likewise document a positive effect of geographical proximity between suppliers and customers (in a business-to-business setting) on supplier innovation. In the retailing context, Walmart tends to open stores in clusters to facilitate knowledge sharing via the easier transfer of experienced managers and other associates from preexisting stores to the new nearby stores (Fettig, 2006). The theoretical operations management (OM) literature argues that the effectiveness of group incentives is contingent upon opportunities to collaborate. Siemsen et al. (2007) analytically show that the optimal design of incentive systems depends upon three linkages among entities (i.e., *outcome*, *help*, and *knowledge*). Thus, we also investigate if the geographical proximity of retail stores moderates the impact of group incentives on store performance.

Furthermore, we examine if the impact of the incentive change on store performance is moderated by the focal store's past performance. The organizational behavior (OB) literature argues that the top performers reduce their performance significantly under the group incentives as their earnings may be reduced by poor performers compared to the store-based incentive scheme (e.g., Honeywell-Johnson & Dickinson, 1999). One possible explanation is that under the group incentive scheme, top-performing stores help nearby bottom-performing ones by utilizing some of their resources. Prior literature (Sandvik et al., 2020) finds "knowledge transfers from highly skilled workers to less skilled ones" (p. 34). Accordingly, we investigate the moderating role of the focal store's past performance.

To answer our research questions, we study store performance using data from 75 stores of a retail chain that changed its incentive plan for store managers from being purely dependent on store performance (we call it the *SP* plan) to being dependent upon both corporate and store performance (we call it the *CPSP* plan). The new *CPSP* plan resembles a differentially divided group incentive (Honeywell-Johnson & Dickinson, 1999), such as gainsharing (Benson & Sajjadiani, 2018) because "part of employee compensation during a particular period is based substantially on the profitability of the company in that period" (Kruse, 1993, p. 5). To examine the causal effect of the incentive change, we further identify a control group from a matched store of another retail chain in the same industry and perform a difference-in-differences (DiD) analysis.

Most literature in OB examines performance implications of group incentives either through lab experiments, surveys, or case studies (see Honeywell-Johnson & Dickinson, 1999, for a review). A handful of papers in economics conduct field experiments (Friebel et al., 2017) or exploit quasi-experimental data (Hamilton et al., 2003) to study group incentives but they have done so at an individual employee level. A few papers have studied the incentive change for store managers (Anderson et al., 2010; DeHoratius & Raman, 2007); however, they do not consider group incentives as well as geographical proximity of stores. Thus, this is one of the first studies to empirically examine the adoption of group incentives for managers and the moderating role of geographical proximity using field data.

We note that our objective is not to test the optimality of the incentive plan or argue broadly about the efficacy of group incentives. Instead, we aim to broaden our understanding of how incentives work in practice, and, more importantly, document how firms can design a more effective incentive scheme for managers by considering stores' geographical network and past performance. This documentation is especially necessary because a large number of incentive plans fail in practice (Kesner et al., 2016). In this regard, this paper resembles a case study similar in scope to prior studies that examined the impact of incentives on performance in field settings (Bandiera et al., 2005; DeHoratius & Raman, 2007).

Our primary findings are as follows: By using a DiD approach along with matched control stores from another retail chain (i.e., control retailer), we find that store performance overall decreases under the new *CPSP* incentive plan. The year–overyear (YOY) sales growth rate is 3.67% smaller under the new *CPSP* plan compared to the old *SP* plan. We further find that the effect of the incentive change is contingent upon (1) the geographical proximity among stores and (2) the focal store's past performance. The effect of the incentive change on store performance is *positively* moderated by the increase in stores within close geographical proximity (i.e., the number of other stores under the same retail chain within 50 miles radius of a focal store) while it is *negatively* moderated by the past performance of the focal store. We perform a number of robustness checks to rule out alternate explanations such as manager selection due to the incentive change (see Section 5.3).

Our paper contributes to the OM literature in the following way. The OM literature has a large body of theoretical research in contracting and incentives, but has a few empirical studies using field data (Cachon, 2003). Notable exceptions include DeHoratius and Raman (2007) who use field data from a retailer to study the impact of incentives on store managers' effort allocation between sales and inventory shrinkage; Akkas and Sahoo (2020) who study how to reduce product waste by designing salesforce incentive scheme; and an emerging stream that examines the effects of contract design on operational performance (Chan et al., 2019; Guajardo et al., 2012; Guajardo, 2018). From a behavioral perspective, Li et al. (2020) compared different types of compensation schemes for store managers using lab experiments. We add to this literature by documenting the impact of group incentives for store managers on store performance, which has not been studied in the prior empirical OM literature.

We demonstrate that the efficacy of incorporating corporate goals into managers' incentive plans is contingent on geographical proximity among stores and past performance of the focal store. A long line of OM literature has recognized that effective retail management requires careful choice of store locations and subsequent management of those locations within the retail network. A number of papers have examined ways to optimize retail locations (see Berman et al., 2007, for a review). Another stream of literature investigates issues around geographically dispersed stores including inter-store learning (Chang & Harrington, 2003) and distribution network (Caro & Gallien, 2010), among others. We contribute to this line of literature by showing how geographical dispersion and past performance can impact the efficacy of group incentives. We show that the optimal incentive structure for retail store managers depends upon both stores' geographical network, as suggested by the OM literature, as well as stores' past performance. Our work is also closely related to the retail operations literature, which studies various issues such as traffic (Mani et al., 2015; Perdikaki et al., 2012), labor (Kesavan et al., 2022), and variety (Lu et al., 2022).

2 | RESEARCH SETTING

We collaborated with a U.S.-based department store retailer with annual revenues exceeding $600 million to obtain our data. More than 5300 associates work on a daily basis in its stores. As a discount department store, this retailer sells high-volume low-margin products: primarily women's, men's, and children's apparel, along with accessories, fragrances, and home furnishing goods. The study period was from February to December 2014. We obtained data from 75 stores for our analysis. We provide details about the data in Section 4.1. Before presenting the structure of the original and new incentive plan, it is useful to describe the nature of store manager activities to better understand the impact of the change in store managers' incentive scheme.

2.1 | Store manager activities

Store managers play pivotal roles in retail organizations. They are responsible for running the store effectively to generate sales. To that end, they need to perform many activities within the store and interact with other teams outside of the store.

The primary day-to-day tasks of managers affecting store sales involve managing/motivating associates and scheduling them to ensure that store operations run smoothly.[3] Store managers motivate and train associates in the store to provide customer service. They also help store associates perform activities that indirectly affect sales such as receiving merchandise, handling customer returns, and operating checkout counter. In addition, scheduling the right number of associates at the right time requires a high amount of manager's effort and significantly affects sales.

In addition to running in-store activities, store managers also need to interact with several teams outside of the store. They work with the human resources department to coordinate hiring strategies for the store. They coordinate promotional activities with the marketing department. They work with merchandising teams to determine the planogram, with replenishment teams to order merchandise, and with logistics teams to schedule deliveries. They also interact with other store managers to share their experience/knowledge around new initiatives or general store execution and, in some cases, to share pooled labor with other stores. These activities require a high level of cooperation among stores and with the corporate office.

While store managers are responsible for day-to-day operations with substantial discretion, they generally work within tight constraints set by the corporate headquarter. The headquarter controls store layout, product selection, pricing, and inventory levels (Garvin & Levesque, 2008; Ton, 2012). Store managers in our setting also do not control marketing dollars, pricing flexibility, and other levers that determine the store performance as these are made centrally.

A store manager's performance and conformance are monitored by a district manager who supervises 13–15 stores in the region. District managers monitor several vital metrics such as sales on a daily and weekly basis. District managers discuss daily performance with store managers and host multiple conference calls every week with all store managers in the region to discuss performance. They also visit stores regularly for close monitoring.

Overall, store managers have little control over what they must do but have considerable discretion over how to assign, sequence, and accomplish tasks. They can have a significant influence on their store performance through improved in-store activities and on other stores' performance through greater cooperation with other stores (Garvin & Levesque, 2008).

2.2 | Incentive plan for store managers

Across the retail industry, store managers' bonus pay is tied either to only their store performance or both their store and parent group (division, region, or corporate) performance. Significant differences are present in the components used to measure store performance across retail organizations. In some organizations, the store performance is measured based on sales and profit goals (van Donselaar et al., 2010); others are based on sales and a salient component of store expense that is under managers' control such as shrinkage (DeHoratius & Raman, 2007) or labor (Terborg & Ungson, 1985). We next provide specific details of the incentive change for the retailer in our study.

In the first two quarters of 2014, store managers were paid quarterly bonuses based on two components of store performance. First, the YOY sales growth was computed for each store in that quarter. Managers of stores with a positive growth rate received 1% of their salary as a bonus for every 1% YOY growth. Second, the deviation of actual labor from the ex post labor target (from the software) was calculated. When actual labor is more than the ex post labor target (i.e., overstaffing), managers at our site are assumed to have spent too much on labor. For every 0.1% negative deviation between the ex post labor target and actual labor (in

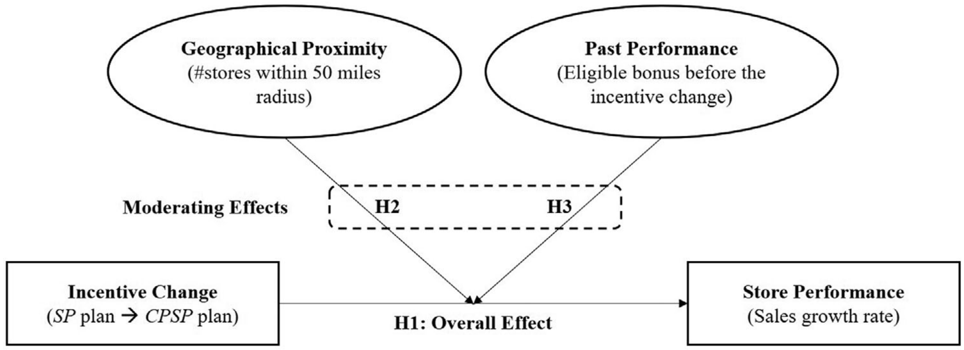

FIGURE 1 Conceptual diagram

percentages of actual sales), the bonus calculated in the first step is reduced by 10%.

In the second half of 2014, the company added a corporate-level target, positive EBITDA (earnings before interest, tax, depreciation, and amortization), as a contingency for bonus eligibility. If the company's EBITDA is not positive, then managers are automatically disqualified from earning a bonus irrespective of store performance. So, the incentive change can be described as a change from a financial incentive plan tied to store performance only (*SP* plan) to one with mixed goals that combine corporate and store performances (*CPSP* plan). We investigate the impact of a change in the incentive plan on store performance.

2.3 | Company's motivation for the incentive change

This retailer identified omnichannel fulfillment as a strategic initiative to compete with online retailers and was planning to offer services such as buy-online-pickup-in-store and buy-online-return-in-store in the near future. This retailer believed that moving into omnichannel retailing would require greater cooperation and coordination among stores as well as with the corporate office (Rigby, 2011). So, senior management decided to change the incentive structure for store managers by tying their bonuses to organizational performance. Such a change would be consistent with theoretical and empirical literature that has argued for introducing group incentives to improve cooperation and coordination among agents. For example, Itoh (1992) terms "induced cooperation" as cooperation among agents induced by a group incentive and Kruse et al. (2010) indicate increased cooperation as an important reason for a group incentive (e.g., profit sharing): "profit/gain sharing employees are more likely to work in teams, to be able to observe co-worker performance, and to have low levels of supervision" (p. 59).

In practice, we observe different ways in which corporate performance is tied to store manager performance. One common approach is to tie part of a manager's compensation to organizational performance and the other to store performance. For example, Kroger ties 40% of store manager compensation to corporate performance and the remaining to their store performance. One disadvantage of this type of incentive structure is that it can lead to the free-riding problem (Holmstrom, 1982), a major obstacle in implementing group incentives. The retailer in our study, on the contrary, overcomes the free-riding problem, as managers cannot earn any bonus if their store performs poorly even if the organization is profitable.

3 | HYPOTHESES DEVELOPMENT

We develop hypotheses concerning the sensitivity of the sales growth rate to the incentive change. We examine the YOY sales growth rate because it is closely related to the incentive scheme for store managers. In our setting, the sales growth rate is the major driver of store managers' bonus as failure to increase the YOY sales growth rate will result in zero bonus even if a manager outperforms in other dimensions of store performance. We first investigate the overall impact of the incentive change on stores' sales growth rate and then the moderating effects of geographical proximity among stores and past performance of the focal store. Our predictions are guided by studies in the OM, economics, and OB literature. Figure 1 provides the conceptual diagram for our hypotheses.

3.1 | Overall effect of the incentive change on store performance

3.1.1 | Positive effect

We examine the arguments behind the positive impact of the incentive change on store performance that motivated this retailer to change its incentive scheme. Tying managers' pay to corporate goals can improve store performance, as group incentives have been known to drive increased monitoring and peer pressure. According to agency theory (Levine & Tyson, 1990), each agent turns into a principal who *monitors* other agents, as agents' pay ties to group incentives, resulting in higher store performance. Kandel and Lazear (1992) theoretically show that *peer pressure* can result in higher productivity of individuals in the team. The new *CPSP* plan could increase both monitoring and peer pressure among store managers. As a part of the multiunit enterprise (Garvin & Levesque, 2008), retail stores in our setting are grouped under a district manager. Store managers report to district managers and discuss their performance through multiple weekly conference calls where all other store managers in the same district attend. Also, performance data are tracked on a daily basis and shared with all store managers. Specifically, the variance of the actual sales from forecasts is shared with other store managers within a district. This could lead to higher peer pressure under the new incentive scheme.

Performance could also improve because of greater cooperation among store managers (Itoh, 1992). In general, the role of cooperative actions such as helping and knowledge sharing in improving organizational performance has been widely acknowledged (D. Katz & Kahn, 1966). Gupta and Govindarajan (2000) document that Nucor Steel's group incentive structure encouraged plant managers to share their own best practices with other plant managers, resulting in overall superior productivity. Store managers could also share their knowledge around best practices to increase store sales through multiple channels including weekly conference calls. Accordingly, we hypothesize the following:

Hypothesis 1a. *The adoption of the CPSP plan will lead to an* increase *in the sales growth rate.*

3.1.2 | Negative effect

Conversely, there are many reasons for performance to decline under the new incentive structure. Increasing the importance of corporate performance naturally results in a reduction in the importance of store performance in managers' incentive plans. This could lead to lower motivation (so lower effort) if store managers perceive they have less control over their pay (Lawler, 1990) because group incentives decrease "the extent to which the worker can influence his or her own pay" (Bucklin & Dickinson, 2001, p. 50). Expectancy theory predicts lower motivation for agents when they cannot see how their efforts translate into outcomes (Vroom, 1964).

The loss of motivation due to less control can play a significant role when the group is large. Compared to the individual incentive, the group incentive increases a plan's "line of sight," which refers to the extent to which an employee feels that s/he can actually influence results, and hence pay (Lawler, 1990). Honeywell-Johnson et al. (1997) note that "as the group size increases, the capacity of an individual worker to control his or her wages under group incentive conditions decreases" (p. 262). Blinder (1990) refers to this as the "1/n*th* problem," in which "*n*" denotes the number of members in the group. According to Blinder, individual performance is likely to decrease as "*n*" increases. Two early field studies find supportive evidence by observing an inverse relationship between group size and performance (Campbell, 1952; Marriott, 1949). As the *CPSP* plan consists of 75 store managers (considered as large in this literature), the incentive change can lead to lower performance due to intensified loss of motivation.

The OB literature draws a further distinction among different types of group incentives by calling those incentives that combine individual and group targets at mixed goal incentives. This literature argues that mixed goals could lead to lower performance compared to purely individual or group incentives. Libby and Thorne (2009) find that mixed goals send confusing signals to the participants about where to direct their effort. Mixed goals can also cause goal conflict and sacrifice effort towards attaining one goal to meet the other goal (Ambrose & Kulik, 1999). For example, Locke et al. (1994) find that faculty members experience goal conflict due to mixed goals around teaching and research, leading to lower research productivity. Finally, mixed goals may split managers' attention across store-specific and group activities, so they may not be able to fully develop collective processes and strategies that are necessary to improve overall group performance (Wageman, 1995). So, store performance could decline under the *CPSP* plan due to several reasons including confusion, goal conflict, and division of attention.

In summary, loss of motivation, especially due to large group size, and issues around mixed goals could make store managers exert less effort on sales-generating activities. Accordingly, we hypothesize the following:

Hypothesis 1b. *The adoption of the CPSP plan will lead to a* decrease *in the sales growth rate.*

3.2 | Moderating factors

We next study conditions under which the adoption of the *CPSP* plan is more effective. Grounded in prior literature, as shown below, we particularly examine two factors—stores' geographical proximity and past performance of the focal store—that moderate the impact of the incentive change on store performance.

3.2.1 | Geographical proximity

We conjecture that geographical proximity among stores can moderate the effect of the incentive change on store sales performance for the following reasons. Prior literature finds that proximity facilitates knowledge transfer. For example, by investigating intranational university–university research collaboration, J. Katz (1994) demonstrates that research collaboration reduces substantially with the distance between the collaborative partners. In the automobile supply chain context, Bray et al. (2019) find higher defect rates when the supply chain is geographically dispersed. Likewise, in a business-to-business setting, Chu et al. (2019) also document a positive effect of geographical proximity between suppliers and customers on supplier innovation partly "because short distance facilitates soft information exchange between two parties" (p. 2445). In the multiunit enterprise context such as retailing, Chang and Harrington (2003) state that "One source of competitive advantage is their superior capacity for learning through the transfer of knowledge among internal units" (p. 541). In practice, Walmart tends to open stores in clusters to facilitate knowledge sharing via the easier transfer of experienced managers and other associates from preexisting stores to the new nearby stores (Fettig, 2006).

In the OM literature, theoretical papers predict that the performance of incentive structures is contingent upon opportunities to collaborate. For example, by identifying three distinct types of linkages in a group (i.e., *outcome*, *help*, and *knowledge*), Siemsen et al. (2007) analytically show that the optimal design of incentive systems depends upon the extent of the three linkages. Retail stores can have these three linkages providing store managers opportunities to collaborate. When a store manager does not receive merchandise on time, s/he affects the delivery for the next store (i.e., *outcome* linkage). Store managers can help other stores by sharing their human resources through labor pooling (i.e., *help* linkage). They can also share their knowledge with others through multiple channels such as conference calls within the same district and manager training program with "sister" stores (i.e., *knowledge* linkage). We conjecture that stores in close geographical proximity have tighter linkages when compared to stores that are far away.

Grounded in the above logic, we expect geographical proximity among stores to moderate the effect of group incentives on store performance. Specifically, if a store has many geographically neighboring stores, the adoption of *CPSP* could have more room for positive effects on performance due to tighter linkages. In contrast, if a store is geographically isolated, the adoption of *CPSP* could have limited room for positive effects due to weaker linkages. Since the overall impact of the *CPSP* plan could be either positive or negative, as argued in the previous hypotheses, we offer the following prediction:

Hypothesis 2. *The effect of the CPSP plan on sales growth rate will be moderated by geographical proximity among stores in a way that stores with more neighboring stores will benefit more from the adoption of the CPSP plan.*

3.2.2 | Past performance

We also conjecture that stores' past performance can moderate the effect of the incentive change on the sales growth rate. When the incentive plan for managers is switched from individual based to group based, the top performers reduce their performance while poor performers continue to perform below average (Honeywell-Johnson & Dickinson, 1999). Previous studies have documented that the amount earned influences the participant's preference and satisfaction with different incentive structures (Bucklin & Dickinson, 2001; Honeywell-Johnson et al., 1997). High performers are likely to earn less money under the group-based incentive as their earnings can be reduced by poor performers (Honeywell-Johnson et al., 2002). Thus, store managers who performed well under the old *SP* plan may reduce their effort after the adoption of *CPSP* because of reduced expected earnings whereas store managers who did not perform well before the incentive change may not suffer from such a lack of motivation.

In addition, it is likely that better-performing stores spillover their resources towards helping bottom-performing ones. Prior research finds a similar mechanism: by conducting a field experiment in a sales call center, Sandvik et al. (2020) identified stars as agents with pretreatment productivity above the median and concluded that "star agents provided knowledge to non-stars" (p. 24). So, we expect bottom-performing stores in general to benefit more from the incentive change as they receive help from other better-performing ones. Accordingly, we hypothesize the following:

Hypothesis 3. *The effect of the CPSP plan on sales growth rate will be moderated by the focal store's past performance in a way that bottom-performing stores in the past will benefit more from the adoption of the CPSP plan.*

4 | DATA AND IDENTIFICATION STRATEGY

4.1 | Data description

4.1.1 | Treatment group

Our main data are obtained from a U.S.-based department store retailer described in Section 2. Data from all 75 stores of this retail chain were collected for the study period (February–December 2014). These data include weekly actual and forecasted sales data. The corporate sales forecasts were generated by a top-down approach using the expected aggregate firm-level sales growth as the primary input. We use the address of each store to calculate the geographical proximity among stores. In addition, we have sales from the previous year (2013) to calculate YOY sales growth. We also have labor information such as actual labor used in the stores as well as the ex post labor target, which is the amount of labor that the store should have used based on the actual sales that were realized.

4.1.2 | Control group

Direct comparison of the performance before and after the incentive change can potentially yield biased estimates due to other confounding effects such as unaccounted for time trends or seasonality in the data. To control for these confounding effects, we identify a control group that is likely to have the same trend and seasonality factors as our focus retailer but is unaffected by the incentive change. We can then draw inferences based on the difference in two differences: the before-and-after difference in performance in stores of the treatment group and the before-and-after difference in performance in stores of the control group.

Our control group was chosen from over 300 stores of another department store retailer (i.e., control retailer). This department store has the same departments as our focus retailer (i.e., treatment retailer): men's, women's, children's apparel, cosmetics, and accessories. The stores of both retailers are not collocated as they are often in different cities. Only five stores are in the same cities and 45 treatment stores had a control store within 100 miles radius. The incentives for store managers of the control retailer, which stayed the same during the study period, are primarily based on sales growth and deviation of actual labor from the ex post labor target. From this control retailer, we obtain actual labor ($) and ex post labor target ($) for each month of 2013 and 2014. As the data from this control retailer were at a monthly level, we aggregated the data from the treatment retailer and performed our primary analysis at the monthly level.[4]

4.2 | Identification strategy

4.2.1 | Difference-in-differences design

We adopt a DiD approach that has been used widely on experimental and quasi-experimental data (Angrist & Pischke, 2009). This approach has been deployed in many OM papers (e.g., Bell et al., 2018; H. S. Lee et al., 2021; J. Lee et al., 2021). Our objective is to examine the impact of the incentive change on the sales growth rate.

Our DiD model takes the following form:

$$Y_{im} = \alpha_i + \beta_1 Treatment_i + \beta_2 After_m + \beta_3 Treatment_i \times After_m + W_m + \mathbf{Z_{im}} + \varepsilon_{im}, \tag{1}$$

where the subscript i denotes store and m denotes month. Y_{im} represents the sales growth rate (i.e., $Sales\_growth_{im}$), measured as the difference between sales in 2014 and those in 2013 in percentages of sales in 2013 ($\frac{Sales - Sales\_last\_year}{Sales\_last\_year}$). The binary variable $Treatment_i = 1$ if a store i belongs to the retailer that changed an incentive for store managers and 0 otherwise. It is used to capture the difference in performance across treatment and control retailers. The binary variable $After_m = 1$ if month m corresponds to the posttreatment period (i.e., after the incentive change) and 0 otherwise. It captures the factors that affect store performance in the period after the incentive change. The parameter of interest is that of the interaction term, that is, β_3, which measures the impact of the incentive change on sales growth. Our preferred specification includes store fixed effects α_i to capture location-specific, time-invariant factors, as well as monthly seasonality W_m (one dummy for each month) to capture time-specific, location-invariant factors. After including them, the coefficients for $Treatment_i$ and $After_m$ are unidentifiable because the former absorb the effects of the latter. However, the store fixed effects and monthly seasonality serve the same purpose as the main effects $Treatment_i$ and $After_m$ in this regression.[5] Therefore, we exclude the main terms and keep the interaction term ($Treatment_i \times After_m$) only for clarity of exposition.

We further include two additional control variables represented in the control vector $\mathbf{Z_{im}}$. First, we use the sales growth rate of the U.S. department stores (*SG_Dept_stores*), obtained from the U.S. Census, to account for macroeconomic and industry-wide effects (Anderson et al., 2010; DeHoratius & Raman, 2007). The Census has monthly-level sales for the entire retail sector and breaks it down to a specific industry level. We obtain monthly-level sales data for all stores under NAICS (North American Industry Classification System) code 4521 (i.e., U.S. department stores). Second, we control for the expected sales growth defined as the deviation between the corporate sales forecast and sales in the same month in the previous year normalized by these prior sales (i.e., $Expected\_SG = \frac{Corporate\_sales\_forecast - Sales\_last\_year}{Sales\_last\_year}$). This captures predictable firm-specific factors as the corporate sales forecast is known to contain firm-specific information (Hassell et al., 1988).

Finally, the disturbance term ε_{im} captures unobservable determinants of operational outcomes at the store-month level. Operational outcomes such as store sales within the same store are unlikely to be independent, as each store faces similar local economic and labor market conditions month by month. We account for this effect by clustering standard errors at the store level in all regressions.

Table 1 provides descriptive statistics of variables during the study period, and Table 2 contains Pearson correlation coefficients among all variables used in our analysis.

To investigate the moderating effect of geographical proximity among stores, we use the store address to compute the distance in miles between all pairs of stores using Google Maps' application programming interface, which accounts for the actual road network conditions. We then count the number of other stores under the same retail chain within 50, 75 and 100 miles radius. Table 3 provides this store network information for the treatment retailer. We use the number of stores within 50 miles radius (*Within*50) in the main model. As a robustness check, we use 75 and 100 miles radius as an alternate cutoff and obtain consistent results.

To explore the moderating effect of stores' past performance, we follow the bonus calculation of the treatment retailer under the *SP* plan to measure the past performance of stores. Specifically, we calculate quarterly sales growth and then apply a 10% penalty for 0.1% of over budget in labor (i.e., $(\frac{Sales - Sales\_last\_year}{Sales\_last\_year}) \times [1 - 100 \times (\frac{Actual\_labor - Labor\_target}{Sales})^{+}]$). We note that if sales growth is

TABLE 1 Descriptive statistics of the variables

Monthly-level (N = 730)	Mean	SD	Min	Median	Max
Sales growth rate	−0.0575	0.0655	−0.197	−0.0623	0.163
Expected sales growth rate	0.0133	0.0595	−0.163	0.0155	0.206
U.S. department stores' sales growth rate	−0.0114	0.0112	−0.0295	−0.0107	0.00766

TABLE 2 Pearson correlation coefficients

	(1)	(2)	(3)
(1) Sales growth rate	1.00		
(2) Expected sales growth rate	**0.52**	1.00	
(3) U.S. department stores' sales growth rate	−0.08	−0.08	1.00

Note: Bold denotes statistical significance at the 1% level.

not positive, then the eligible bonus for a store manager is zero; however, we still compute the amount (possibly negative) from the bonus calculation for each store to keep track of how far away the store was from getting a bonus. Among 75 stores, 2 stores received bonuses for both quarters under the *SP* plan, 25 stores received bonus once, and 48 stores did not receive any bonus during two quarters before the incentive change. We then calculate the average of this bonus across two quarters and use it (*Bonus*) to capture the moderating effect of the focal store's past performance.

Using the above two moderators, we extend our DiD model as follows:

$$\begin{aligned} Y_{im} = {} & \alpha_i + \beta_1 Treatment_i + \beta_2 After_m + \beta_3 Treatment_i \times After_m \\ & + \beta_4 Within50_i + \beta_5 Treatment_i \times Within50_i \\ & + \beta_6 After_m \times Within50_i \\ & + \beta_7 Bonus_i + \beta_8 Treatment_i \times Bonus_i \\ & + \beta_9 After_m \times Bonus_i \\ & + \beta_{10} Treatment_i \times After_m \times Within50_i \\ & + \beta_{11} Treatment_i \times After_m \times Bonus_i \\ & + W_m + \mathbf{Z_i} + \varepsilon_{im}. \end{aligned} \tag{2}$$

It is necessary to include all the lower order interaction terms in the model as shown in Equation (2). Again, our preferred specification includes store fixed effects (α_i) and monthly seasonality (W_m), which absorb the effects of lower order interaction terms. Thus, we only report the variables of interest ($Treatment_i \times After_m$, $Treatment_i \times After_m \times Within50_i$, and $Treatment_i \times After_m \times Bonus_i$) for clarity of exposition.[6]

4.2.2 | Key identifying assumption for DiD

The validity of the DiD approach hinges upon the existence of the *parallel trend*. It does not require the control and treatment to be a priori the same in all aspects but requires that the two groups follow the same trend in the absence of the treatment, while the levels may differ due to differences in characteristics.[7] For the parallel trend assumption, we compared the treatment and control groups' trends in the sales growth rate by estimating the following model using data from the pretreatment period as used in the literature (Fisher et al., 2019):

$$\begin{aligned} Y_{im} = {} & \alpha_i + \gamma_1 Trend_m + \gamma_2 Treatment_i \times Trend_m \\ & + W_m + \mathbf{Z_i} + \varepsilon_{im}, \end{aligned} \tag{3}$$

where $Trend_m$ denotes the number of months since the start of the observation period (i.e., February 2014). Using all stores from the control retailer resulted in significant γ_2 ($p < 0.01$), indicating a potential violation of the parallel trend assumption.

4.2.3 | Developing a matched sample

Having determined that the DiD model cannot be executed on the entire sample of control stores, we proceeded to pick appropriate control groups that are similar to the treatment group based on matching. Matching is widely used to construct a quasi-control group based on similar characteristics to those of the treatment group (Heckman et al., 1998). Matching along with DiD can be especially powerful to obtain causal results as matching produces a control group that is similar in observable characteristics, while DiD

TABLE 3 Store network (treatment retailer)

	Number of other stores within "X" miles radius						
"X" =	0	1	2	3	4	5	6
50	24 (32%)	19 (25.33%)	11 (14.67%)	5 (6.67%)	12 (16%)	4 (5.3%)	–
75	15 (20%)	17 (22.67%)	15 (20%)	7 (9.33%)	8 (10.67%)	10 (13.33%)	3 (4%)
100	4 (5.33%)	17 (22.67%)	16 (21.33%)	6 (8%)	15 (20%)	7 (9.33%)	10 (13.33%)

Note: The retail chain has 75 stores in total during the study period.

permits time-invariant unobservable differences between the treatment and control groups. Propensity score matching to develop a matched control group on which to estimate treatment effects with a DiD specification using panel data has been used in many papers in OM (e.g., Levine & Toffel, 2010) for program evaluations. So, we use propensity score matching to pick our control group (Rosenbaum & Rubin, 1983).

Given our limited sample size of around 300 stores, we carefully choose the variables to perform matching as too many variables will reduce the control group size and too little may produce a dissimilar control group. We consider factors related to stores' historical performance and their geographical market characteristics. For the former, we seek to capture a store's multidimensional performance, using trends in sales growth, the accuracy in labor scheduling (both with and without taking its absolute value), and labor productivity. We only use pretreatment data for matching to ensure that the matching variables are unaffected by the treatment. We include trends in store performance to ensure that the parallel trend assumption is not violated. We further add sales growth as matching criteria because it is possible that stores with positive growth might be managed differently from stores with negative growth. For market characteristics, we use median household income ($), number of households, and retail market size (i.e., retail sales), obtained from the Census data using the store's zip code.[8] In summary, we perform matching based on eight covariates. Our assumption is that once matched on the covariates the difference in performance we observe between the treatment and control groups is only due to the incentive change. Table A.1 in the Supporting Information Appendix shows the logit results that generated propensity scores. Using probit provides similar results.

We implement the nearest neighbor matching with replacement as it has been found to be suitable when the distributions of treatment stores' and control stores' propensity scores differ (Heckman et al., 1997). This yielded a match for 75 stores for the treatment retailer, using 52 control stores from around 300 stores for the control retailer. As a robustness check, we employ the propensity score weighting method where the entire sample of treatment and control stores are analyzed.

With the control group generated using the matching process, we perform a number of tests to ensure the validity of the matched control group. First, we test the balance of the covariates in the treatment and control groups. Table A.2 in the Supporting Information Appendix shows the results of the t-tests of the eight characteristics that we used to construct the matches. We find all eight differences to be statistically indistinguishable from zero at the 1% level. Second, we test the parallel trend assumption using Equation (3). As shown in Column (2) of Table 6, we do not find evidence of a violation in the parallel trend. Finally, we perform a placebo test where we assume that the intervention occurred at the end of the first quarter and test for any significant trend between the first 3 months and the second 3 months prior to the incentive change. This test also serves as a robustness check for the DiD specification, and as we show later (Column (3) of Table 6) we do not observe any difference indicating that there was no trend in the sales growth rate before the actual intervention at the end of the second quarter.

5 | RESULTS: EFFECT OF THE INCENTIVE CHANGE ON STORE PERFORMANCE

In this section, we present the estimated effects of the incentive change from the *SP* plan to the *CPSP* plan on the sales growth rate. As we use the nearest neighbor matching with replacement, all regressions are estimated by weighted least squares using the frequency weight (i.e., the frequency with which the observation is used as a match). We first show the overall effect, followed by moderating effects of stores' geographical proximity and past performance of the focal store. Finally, we conduct multiple robustness analyses to demonstrate that our results are consistent under alternative explanations.

5.1 | Overall effect

We run Equation (1) to test Hypotheses 1A and 1B, and Table 4 provides the results. In Column (1), we observe a significant impact of the incentive change on sales. The estimated coefficient of $Treatment \times After$ is negative and statistically significant ($\beta_3 = -0.0367$, $p < 0.001$), indicating on average a 3.67% smaller sales growth rate under the new *CPSP* plan than under the old *SP* plan, keeping everything else equal. This difference is substantial, as the average monthly sales growth rate of the treatment retailer after the incentive change is -7.67%.[9] So, the new incentive plan has driven 47.85% ($= 3.67/7.67 \times 100$) worsening of sales performance by store managers. The result shows that store managers generate smaller sales growth under the new *CPSP* plan, supporting H1B.

We find that the estimated coefficients of the control variables are in the expected direction. The expected sales growth is positively associated with the sales growth rate. This confirms that the corporate forecast is a good predictor for the sales, consistent with prior literature that argues for the value of corporate forecast due to its embedded firm-specific information (e.g., Hassell et al., 1988). The sales growth of the U.S. department store is also positively associated with store sales growth. This is again consistent with prior literature (Anderson et al., 2010; DeHoratius & Raman, 2007), pointing out the importance of controlling for the industry trend.

5.2 | Moderating effects

We run Equation (2) to test Hypotheses 2 and 3. We first check the moderating effects separately. Column (2) includes the moderator based on stores' geographical proximity and

TABLE 4 Impact of the incentive change on sales growth (DiD)

	Monthly *Sales_growth*			
	Overall effect	Moderating effects		
	(1)	(2)	(3)	(4)
Dependent variable	H1	H2	H3	H2 and H3
Treatment × After	−0.0367***	−0.0475***	−0.0515***	−0.0707***
	(0.0064)	(0.00822)	(0.00906)	(0.0107)
Treatment × After × Within50		0.00587*		0.00848***
		(0.00263)		(0.00242)
Treatment × After × Bonus			−0.340*	−0.435**
			(0.143)	(0.137)
Expected_SG	0.311***	0.316***	0.312***	0.320***
	(0.0336)	(0.0335)	(0.0354)	(0.0351)
SG_Dept_stores	2.579***	2.569***	2.553***	2.533***
	(0.451)	(0.452)	(0.450)	(0.452)
Store FEs	Y	Y	Y	Y
Month FEs	Y	Y	Y	Y
Observations	1,462	1,462	1,462	1,462
Adjusted R^2	0.521	0.523	0.525	0.529

Note: Standard errors, clustered by store, are reported in parentheses. We have included single, double, and triple interactions in the specification. For simplicity, we only report the variables of interest. We report the full results including the coefficient estimate for both store and month fixed effects in Table A.4 in the Supporting Information Appendix.
Abbreviations: FEs, fixed effects; Y, Yes.
$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$.

Column (3) contains the moderator based on a focal store's past performance. Then, we include both moderators in one model (Column (4)). All results are consistent. We interpret our results based on Column (4) because of the concern about biased coefficient estimates (when estimated separately) due to the link between the two moderators.[10]

We first examine the moderating effect of geographical proximity among stores. As illustrated in Column (4), the coefficient estimate of *Treatment* × *After* × *Within*50 is positive and statistically significant ($\beta_{10} = 0.00848$, $p < 0.001$), indicating that the effect of the incentive change on the sales growth rate is *positively* moderated by the number of stores within 50 miles radius. As shown in Figure 2a, with every incremental store within 50 miles radius, the negative impact of the incentive change decreases significantly. This result provides empirical validation for the prediction from the theoretical OM literature (Siemsen et al., 2007) that the effect of incentives is contingent on the extent of the linkages among entities, supporting H2. The presence of heterogeneous effects alleviates concerns about unobservable firm-wide shocks occurring concurrently with the incentive change driving our findings from H1B.

We also examine whether the past performance of the focal store moderates the impact of the incentive change on the performance. As shown in Column (4), the coefficient estimate of *Treatment* × *After* × *Bonus* is negative and statistically significant ($\beta_{11} = -0.435$, $p < 0.01$), indicating that the effect of the incentive change on the sales growth rate is *negatively* moderated by the focal store's past performance. As shown in Figure 2b, with a higher eligible bonus amount before the incentive change, the magnitude of the negative impact of the incentive change increases significantly. So, our results indicate that bottom-performing stores suffer less than better-performing stores, which supports H3.

5.3 | Alternative explanations and robustness checks

Using data from a U.S.-based retailer, we attempt to uncover the impact of the incentive change on store performance and the moderating roles of geographical proximity and past performance. In what follows, we validate our results under various settings and address potential alternative explanations.

5.3.1 | Effort versus selection

Economics theory suggests that performance-based incentive contracts can increase an organization's overall performance by (*a*) attracting and retaining more productive employees (i.e., selection effect) and (*b*) inducing employees to increase or to better allocate their effort (i.e., effort effect). In our setting, the introduction of the new *CPSP* plan for store managers could result in high-ability managers leaving the firm as their earnings decline and the new managers could be less competent than the experienced ones, driving store

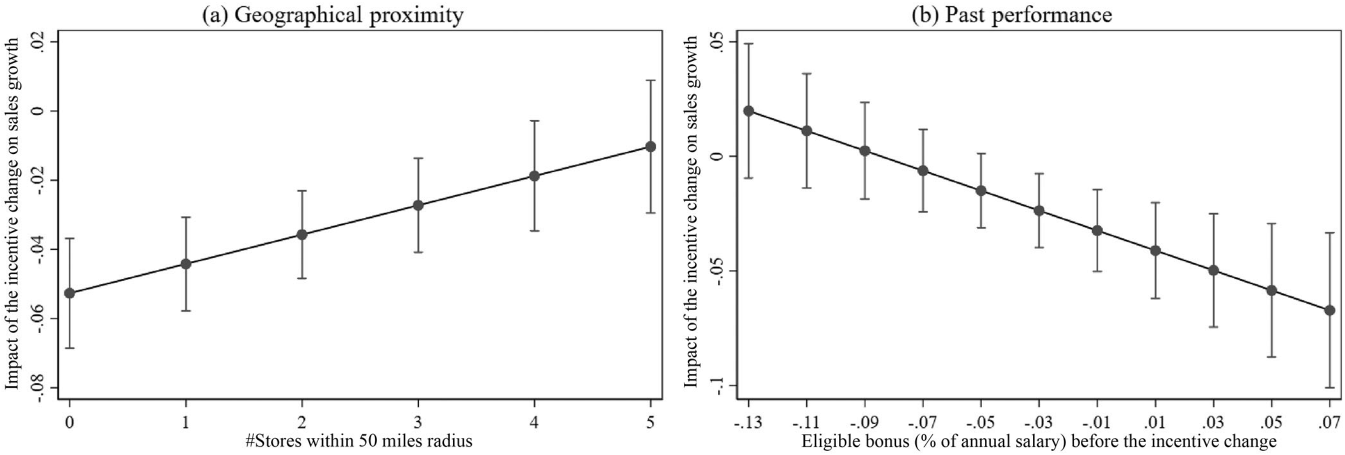

FIGURE 2 Moderating role of geographical proximity and past performance
Note: Figures are based on Column (4) of Table 4. Bars represent the 95% confidence interval.

performance down. Prior research has documented such a selection effect (e.g., Banker et al., 2000; Lazear, 2000).

To rule out this alternative explanation based on the selection mechanism, we perform our DiD analysis with only stores where no managerial turnover occurs during the study period. Among the 75 stores in our sample, 33 stores (i.e., 44%) have no manager turnover in the fiscal year 2014 and our matching process yields 28 matched control stores. We included the turnover of any member of the store management team who can influence store operations, such as the general store manager and assistant store managers. As shown in Table 5 and Figure A.2 in the Supporting Information Appendix, even after removing the selection effect, we continue to observe a significant effect of the change in incentives on sales growth rate and moderating roles of geographical proximity and past performance. So, our main findings are attributable to effort reduction among managers.

5.3.2 | Profit consciousness

An alternative explanation to our findings is that managers might have become more profit conscious when the corporate profitability metric was included in their bonus structure. If so, store managers may focus on selling higher margin products that may drive sales down but potentially increase profits.

We first check whether the gross margin (GM) has increased after the incentive change. From the 10-Q statements for this retailer available from Compustat, we find that the average GM, measured as $(Sales - COGS)/Sales$, in the two quarters following the incentive change is 40.8% while the average GM in the two quarters prior to the incentive change was 43.33%. So, the GM analysis does not support the alternate explanation that managers have become more profit conscious and started selling higher margin products.

We then check store profit. Ideally, we would analyze overall store profitability data to determine if store profits have increased as a result of the new incentive plan. However, we do not possess the data. So we create a new variable, $Imputed\_profit_{im}$ $(= \frac{GM_q \times Sales_{im} - Actual\_labor_{im}}{Sales_{im}})$, where GM_q is the quarterly gross margin. Labor costs constitute the majority of the controllable expense in a retail store, even up to 85% for a large retail chain examined by Kesavan et al. (2014), so our *imputed_profit* metric is likely to be highly correlated with store profits. We check the validity of this metric using store profit data from the control retailer. We use 24 months of data from around 300 stores to compute *imputed_profit* and compare it to a true store profit ratio (i.e., net profit/sales). The within-store correlation is 98%, suggesting the suitability of using *imputed_profit* as a proxy for the store profit. So, we perform the analysis with the *imputed_profit* metric and report the result in Column (1) of Table 6. We find that the *imputed_profit* has significantly reduced after the incentive change for this retailer ($p < 0.001$). Although not conclusive, these results suggest that our findings are less likely to be driven by profit consciousness.

5.3.3 | Placebo test

Although DiD analysis along with propensity score matching on the control retailer significantly mitigates concerns of endogenous change in the incentive scheme, we consider the possibility of an unobserved external shock that led to the observed impact on store performance for the matched treatment stores. We perform a *placebo test* to ensure that the observed impact on store performance for the matched treatment stores is attributable to the incentive change, not an unobserved external shock. We repeat the DiD regression in Equation (1) on data during the pretreatment period. During this old *SP* plan period, we falsely assume that the

TABLE 5 Effort versus selection (33 treated stores and 28 control stores)

	Monthly *Sales_growth*			
	Overall effect	Moderating effects		
	(1)	(2)	(3)	(4)
Dependent variable	H1	H2	H3	H2 and H3
Treatment × *After*	−0.0406***	−0.0603***	−0.0526***	−0.0743***
	(0.00979)	(0.0118)	(0.0131)	(0.0139)
Treatment × *After* × *Within*50		0.0102*		0.0110**
		(0.00439)		(0.00333)
Treatment × *After* × *Bonus*			−0.300	−0.349*
			(0.210)	(0.169)
Expected_SG	0.278***	0.284***	0.273***	0.281***
	(0.0591)	(0.0599)	(0.0584)	(0.0585)
SG_Dept_stores	1.862**	1.845**	1.808**	1.789**
	(0.632)	(0.635)	(0.634)	(0.635)
Store FEs	Y	Y	Y	Y
Month FEs	Y	Y	Y	Y
Observations	759	759	759	759
Adjusted R^2	0.489	0.493	0.495	0.500

Note: Standard errors, clustered by store, are reported in parentheses. We have included single, double, and triple interactions in the specification. For simplicity, we only report the variables of interest.
Abbreviations: FEs, fixed effects; Y, Yes.
$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$.

TABLE 6 Robustness check: Profit consciousness and the validity of DiD design

	Imputed_profit	Monthly *Sales_growth*	
	(1)	(2)	(3)
Dependent variables	Profit consciousness	Parallel trend	Placebo test
Treatment × *After*	−0.0140***		
	(0.00133)		
Trend		−0.0205***	
		(0.00569)	
Treatment × *Trend*		−0.006	
		(0.00349)	
Treatment × *After(false)*			0.00392
			(0.00972)
Expected_SG		0.310***	0.383***
		(0.0836)	(0.0796)
SG_Dept_stores		2.935**	36.97***
		(1.069)	(7.982)
Store FEs	Y	Y	Y
Month FEs	Y	Y	Y
Observations	1,462	736	736
Adjusted R^2	0.919	0.516	0.512

Note: Standard errors, clustered by store, are reported in parentheses.
Abbreviations: FEs, fixed effects; Y, Yes.
$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$.

onset of treatment occurs one quarter before it actually does. If there is an unobservable event during this period that led to the observed store performance being affected in the matched treatment stores, then the estimated coefficient of *Treatment* × *After(false)* should be statistically significant. As seen in Column (3) of Table 6, the estimated treatment effect is statistically indistinguishable from zero. This ensures that the observed change is more likely due to the incentive change, as opposed to some alternative unobserved events.

5.3.4 | Propensity score weighting

One of the disadvantages of the propensity score matching method is that we lose a large amount of data from the control retailer. Alternatively, we can use all stores from the control retailer by using the propensity scores as sampling weights (Hirano et al., 2003), as propensity score weighting reweights treatment and control observations to make the two populations comparable in their observable covariates. Following Hirano and Imbens (2001), we define regression weights ω as follows:

$$\omega(W, x) = \frac{W}{\hat{e}(x)} + \frac{1 - W}{1 - \hat{e}(x)}, \tag{4}$$

where $W = 1$ indicates a treated store and $\hat{e}(x)$x) is the estimated probability of being treated. After obtaining these weights, we estimate our DiD model, including those weights

TABLE 7 Robustness check: DiD with propensity score weighting

	Monthly *Sales_growth*			
	Overall effect	Moderating effects		
	(1)	(2)	(3)	(4)
Dependent variable	**H1**	**H2**	**H3**	**H2 and H3**
Treatment × After	−0.0319***	−0.0406***	−0.0398***	−0.0552***
	(0.00576)	(0.00698)	(0.00787)	(0.00807)
Treatment × After × Within50		0.00720**		0.00902***
		(0.00267)		(0.00254)
Treatment × After × Bonus			−0.173	−0.275*
			(0.123)	(0.106)
Expected_SG	0.360***	0.367***	0.356***	0.366***
	(0.0356)	(0.0340)	(0.0361)	(0.0348)
SG_Dept_stores	2.539***	2.518***	2.519***	2.489***
	(0.379)	(0.381)	(0.377)	(0.379)
Store FEs	Y	Y	Y	Y
Month FEs	Y	Y	Y	Y
Observations	3,623	3,623	3,623	3,623
Adjusted R^2	0.457	0.461	0.461	0.467

Note: Standard errors, clustered by store, are reported in parentheses. We have included single, double, and triple interactions in the specification. For simplicity, we only report the variables of interest.
Abbreviations: FEs, fixed effects; Y, Yes.
$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$.

in the estimation. This technique has been used in several papers in OM (e.g., Bell et al., 2018). We use this technique to retest all hypotheses, and the results are reported in Table 7 and Figure A.3 in the Supporting Information Appendix. The results are all similar to our main results based on the propensity score matching (Table 4 and Figure 2).

5.3.5 | Serial correlation

In the main analysis, we use clustered standard error to address an issue of overestimating t-statistics and significance levels because of potential serial correlation in the error. Here we use an alternate approach. Bertrand et al. (2004) point out that DiD methodology can produce high false positives in the presence of serial correlation. They suggest taking a simple average of the data before and after the treatment and running the regression with a panel of length 2. Table 8 and Figure A.4 in the Supporting Information Appendix show the results. Although this method has a downside due to low power, we still obtain the same results, indicating that the significance of our main results is not driven by serial correlation.

5.3.6 | Alternative measure of geographical proximity

In the main model, we use 50 miles as a threshold because customers are known to travel typically around 20–25 miles from home for regular, yet less frequent purchases such as apparel. Nevertheless, we check the robustness of our results with different cutoffs such as 75 and 100 miles radius. We obtain consistent results when we use 75 miles radius (Columns (1) and (2) in Table 9 and Figure A.5 in the Supporting Information Appendix) and 100 miles radius (Columns (3) and (4) in Table 9 and Figure A.6 in the Supporting Information Appendix), mitigating concerns about the way we operationalize the variable.

5.3.7 | Controlling for demand-side factors

Our geographical proximity measure can be confounded by other demand-side factors. For example, if the economy is improving in urban areas more than in rural areas, then urban stores with greater neighboring stores will likely see higher sales. To mitigate such concern, we do the following. In the main model, we have store fixed effects that can capture any time-invariant, store-varying factors such as differences in local demand across store locations. We also have expected sales growth which can capture the expected demand for the store. Nonetheless, we attempt to directly control for other demand-side factors using Census data (at a zip-code level). Prior literature (e.g., Fisher et al., 2020) used a similar approach to characterize demand aspects of the store. We include (a) median household income (*Med_HHIncome*), (b) population (*Population*), and (c) median age of the population (*Med_Age_Population*) in our model specification,

TABLE 8 Robustness check: DiD with a panel of length 2

	Monthly *Sales_growth*			
	Overall effect	Moderating effects		
	(1)	(2)	(3)	(4)
Dependent variable	H1	H2	H3	H2 and H3
Treatment	−0.0273***	−0.0227**	−0.0146***	−0.0104*
	(0.00590)	(0.00791)	(0.00395)	(0.00492)
After	0.0155*	0.0104	0.0177**	0176*
	(0.00668)	(0.00983)	(0.00645)	(0.00844)
Treatment × *After*	−0.0244***	−0.0322***	−0.0548***	−0.0715***
	(0.00682)	(0.00992)	(0.00803)	(0.0102)
*Within*50		0.000538		0.0000500
		(0.000674)		(0.000256)
Treatment × *Within*50		−0.00111		−0.00212**
		(0.00218)		(0.000719)
After × *Within*50		0.000796		0.0000244
		(0.00107)		(0.000901)
Treatment × *After* × *Within*50		0.00663*		0.00819***
		(0.00270)		(0.00235)
Bonus			0.753***	0.746***
			(0.0500)	(0.0508)
Treatment × *Bonus*			−0.0484	−0.0305
			(0.0524)	(0.0554)
After × *Bonus*			0.0730	0.0780
			(0.120)	(0.120)
Treatment × *After* × *Bonus*			−0.402**	−0.487***
			(0.143)	(0.141)
Expected_SG	0.605***	0.597***	0.312***	0.323***
	(0.0963)	(0.0937)	(0.0599)	(0.0626)
SG_Dept_stores	1.207	0.927	−1.093	−1.164
	(3.168)	(3.095)	(1.562)	(1.608)
Observations	254	254	254	254
Adjusted R^2	0.548	0.562	0.777	0.787

Note: Standard errors, clustered by store, are reported in parentheses.
$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$.

instead of store fixed effects.[11] As shown in Table 10 and Figure A.7 in the Supporting Information Appendix, we still find consistent results, especially the positive moderating effect of geographical proximity among stores, even when controlling for other demand-side factors. This mitigates concerns that geographical proximity is confounded by other demand-side effects.

5.3.8 | Alternative way to account for seasonality

In the main model, we used the DiD specification with two controls (expected sales growth based on the corporate forecast and industry trend) to account for seasonality. We consider an alternative way replacing the corporate forecast as they may suffer from overoptimistic bias.[12] Specifically, we use a longer time series (covering three years, 2012–2014) to account for potential seasonality by including yearly and monthly dummies, instead of controlling for corporate forecast. We note that we have longer time series data for only the treatment retailer, not the control retailer. Thus, we rely on the *first-difference* approach (Roberts & Whited, 2013), which uses the difference before and after the incentive change. Since we do not have sales in 2011, which prevents us from calculating sales growth in 2012, we use sales as a measure of store performance. As shown in Table 11, we find consistent results of worsened store performance after the incentive

TABLE 9 Robustness check: Alternative measure of geographical proximity

	Monthly *Sales_growth*			
	75 miles radius		100 miles radius	
	(1)	(2)	(3)	(4)
Dependent variable	H2	H2 and H3	H2	H2 and H3
Treatment × After	−0.0558***	−0.0804***	−0.0642***	−0.0866***
	(0.00908)	(0.0109)	(0.0101)	(0.0109)
*Treatment × After × Within*75	0.00653*	0.00873***		
	(0.00258)	(0.00245)		
*Treatment × After × Within*100			0.00580*	0.00723**
			(0.00259)	(0.00237)
Treatment × After × Bonus		−0.450**		−0.416**
		(0.134)		(0.131)
Expected_SG	0.314***	0.319***	0.309***	0.312***
	(0.0329)	(0.0347)	(0.0330)	(0.0347)
SG_Dept_stores	2.569***	2.532***	2.581***	2.551***
	(0.452)	(0.452)	(0.450)	(0.450)
Store FEs	Y	Y	Y	Y
Month FEs	Y	Y	Y	Y
Observations	1,462	1,462	1,462	1,462
Adjusted R^2	0.524	0.531	0.525	0.531

Note: Standard errors, clustered by store, are reported in parentheses. We have included single, double, and triple interactions in the specification. For simplicity, we only report the variables of interest.
Abbreviations: FEs, fixed effects; Y, Yes.
$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$.

change. This result based on an alternate way to account for seasonality mitigates the concern about the overoptimistic corporate forecast. We also note that our main identification strategy, DiD, can handle potential confounding effects due to seasonality.

5.3.9 | Supply shocks

If the variability of supply to stores increases after the incentive change either due to supplier issues or internal logistics issues, our estimates can be biased. To mitigate this concern, we use aggregate weekly data of forecasted and actual merchandise delivered to stores from the treatment retailer's distribution centers. To examine weekly supply shocks in each of the stores, we compute the percentage error between *actual* and *forecasted* merchandise delivered to the stores (i.e., *PE_item_received*). Again, the information is available only for the treatment retailer, and we rely on the *first-difference* approach for this analysis. As shown in Table 12, we find that the error in inventory delivered to the stores has not changed after the incentive change regardless of the potential collaboration opportunities ($p > 0.1$). This result rules out the alternate explanation that supply shocks in the second half of the year could have caused performance to decline.

6 | CONCLUSION

The OM literature has a large body of theoretical research in contracting and incentives but there are few empirical studies using data from the field. Cachon (2003) comments that such OM literature "contains a considerable amount of theory, but an embarrassingly paltry amount of empiricism" (p. 30). It is especially important to study the practice of incentives with field data since poorly designed incentives can result in unintended consequences such as damaged company reputations and lowered financial performances (Kesner et al., 2016). Our paper adds to a growing body of the literature on the empirical study of incentives in retailing (Akkas & Sahoo, 2020; DeHoratius & Raman, 2007) and other settings (Chan et al., 2019; Guajardo, 2018; Guajardo et al., 2012) as well as theoretical studies (Siemsen et al., 2007) by documenting the important factors to consider when designing a new incentive plan for store managers.

As many retailers have recently moved towards omnichannel retailing which requires greater cooperation and coordination among stores as well as with the corporate office (Rigby, 2011), understanding the efficacy of group incentives for store managers is imperative for retailers. Our results demonstrate that group incentives resulted in lower sales growth on average in this retail chain, but closer investigation reveals that the efficacy of the new *CPSP* plan is

TABLE 10 Robustness check: Demand-side factors

	Monthly *Sales_growth*			
	Overall effect	Moderating effects		
	(1)	(2)	(3)	(4)
Dependent variable	**H1**	**H2**	**H3**	**H2 and H3**
Treatment × *After*	−0.0264***	−0.0362***	−0.0512***	−0.0705***
	(0.00641)	(0.00835)	(0.00853)	(0.00993)
Treatment × *After* × *Within*50		0.00656*		0.00854***
		(0.00253)		(0.00231)
Treatment × *After* × *Bonus*			−0.342*	−0.436***
			(0.134)	(0.129)
Expected_SG	0.509***	0.511***	0.318***	0.323***
	(0.0491)	(0.0470)	(0.0318)	(0.0313)
Median_HHIncome	0.000000427	0.000000370	0.000000192	0.000000208
	(0.000000341)	(0.000000374)	(0.000000114)	(0.000000121)
Population	2.47e-08*	2.32e-08	2.27e-10	2.69e-10
	(1.13e-08)	(1.18e-08)	(4.56e-09)	(4.40e-09)
Median_Age_Population	0.00130	0.00140	0.000242	0.000142
	(0.000842)	(0.000885)	(0.000340)	(0.000356)
Month FEs	Y	Y	Y	Y
Observations	1,462	1,462	1,462	1,462
Adjusted R^2	0.388	0.390	0.531	0.534

Note: Standard errors, clustered by store, are reported in parentheses. We have included single, double, and triple interactions in the specification. For simplicity, we only report the variables of interest.
Abbreviations: FEs, fixed effects; Y, Yes.
$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$.

TABLE 11 Robustness check: Regression results using 3 years of data (2012–2014)

Dependent variable	**Log of monthly sales (*ln* (*Sales*))**
CPSP(= *Treatment* × *After*)	−0.0177*
	(0.0068)
Ln(*Dept_store_sales*)	0.117
	(0.0805)
Store FEs	Y
Year FEs	Y
Month FEs	Y
Observations	2,415
Adjusted R^2	0.770

Note: Standard errors, clustered by store, are reported in parentheses.
Abbreviations: FEs, fixed effects; Y, Yes.
$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$.

TABLE 12 Merchandise delivered to the stores

	PE_item_received	
	(1)	(2)
Dependent variable		**Geographical proximity**
CPSP(= *Treatment* × *After*)	−0.11	0.17
	(0.26)	(0.15)
CPSP × *Within*50		−0.17
		(0.14)
Store FEs	Y	Y
Observations	2,524	2,524
Adjusted R^2	0.030	0.030

Note: Standard errors, clustered by store, are reported in parentheses.
Abbreviations: FEs, fixed effects; Y, Yes.
$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$.

contingent on multiple important factors, such as a geographical store network and past performance of stores. These findings suggest that retailers need to consider their store network, to understand possible collaboration opportunities, and stores' past performance, to understand their potential loss of motivation, when they design incentives for store managers. Prior research finds that the loss of motivation increases with group size (Honeywell-Johnson et al., 1997). So, future research may examine if tying store manager's incentives to district-level performance, rather than corporate performance, alleviates some of the loss of motivation for top performers.

Our study shares a limitation similar to many empirical studies in OM in the past (e.g., DeHoratius & Raman, 2007;

van Donselaar et al., 2010) that examined only one firm, but doing so allows us to obtain cleaner results using detailed data. Nevertheless, we exercise caution when generalizing our findings because a wide variety of group incentives exist in practice (Nalbantian & Schotter, 1997). Another limitation is that we examine a firm-wide incentive change for store managers, similar to prior literature (Anderson et al., 2010; DeHoratius & Raman, 2007), bringing the concern of unobservable firm-wide shocks. We believe that such concern, while present, is low in our case because we show our results based on DiD with a matched control store from another retailer in the same industry. Nevertheless, by randomizing incentives across managers in field experiments, future research may be able to directly address this issue. Finally, our approach to account for demand-side factors via Census data (e.g., median household income) has been used in the literature (e.g., Fisher et al., 2020), but is not perfect, opening opportunities for the future study. We hope our study sheds light on the impact of incentives on performance and the underlying factors driving the impact to draw greater attention from OM scholars to study the practice of incentives in greater detail using field data.

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ENDNOTES

[1] See "Compensation Discussion and Analysis" (https://s1.q4cdn.com/137099145/files/doc_downloads/irw/governance_internal_page_documents/Compensation.pdf)

[2] See "Corporate Management Bonus Plan - The Home Depot Inc." (http://corporate.findlaw.com/contracts/compensation/corporate-management-bonus-plan-the-home-depot-inc.html)

[3] In some large stores, department managers might plan the labor. The retail chain in our study does not have department managers, and store managers make labor scheduling decisions.

[4] We drop December when we aggregate data at a monthly level because we do not have complete month information (i.e., the information for the last 4 days of December is not available).

[5] We also performed a DiD model without the two fixed effects while including the two direct terms ($Treatment_i$ and $After_m$) and obtained consistent results as shown in Table A.3 and Figure A.1 in the Supporting Information Appendix.

[6] We report the results without the two fixed effects while including all the lower order terms in Table A.3 in the Supporting Information Appendix. We obtain consistent results.

[7] "Threats to the internal validity of the DD estimator cannot come from either permanent differences between the treatment and control groups, or shared trends" (Roberts & Whited, 2013, p. 525).

[8] Prior literature (e.g., Fisher et al., 2020) used a similar approach.

[9] The average monthly sales growth rate of the treatment retailer before the incentive change is −4.49%.

[10] We appreciate an anonymous reviewer for pointing this out.

[11] As the zip-code level demand characteristics do not vary across time, we are unable to include both store fixed effects and demand-side factors in one model. *SG_Dept_stores* is also omitted.

[12] We appreciate an anonymous reviewer for pointing this out.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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